حافظ الغذاء

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Guardian of the food supply

TECHNICAL POSITION PAPER

ورقة موقف تقنية

One Engine, **Six Jurisdictions**

Harmonizing food-safety AI across the GCC without flattening national law

A reference architecture in which one shared decision engine serves the United Arab Emirates, Saudi Arabia, Qatar, Kuwait, Oman and Bahrain at the same time, and every decision it issues arrives carrying the national instrument that authorised it.

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EXECUTIVE SUMMARY

The case in one page

The Gulf imports up to **85 per cent of the food it eats**. Almost all of it arrives cold, moves fast, and enters a legal regime that is genuinely different in each of the six states. That combination is the exposure.

Detection is not the hard part. Sensors already tell an operator when a reefer has warmed up. The hard part is the next thirty minutes: deciding what the load is now, who has to be told, and whether anyone can defend that call later.

Today that judgement rests with individual inspectors and quality officers reading regulations in PDF form. It does not scale with import volume, it is not consistent between two ports in the same country, and it leaves behind almost nothing an authority could put in front of a tribunal.

REFERENCE BUILD, MEASURED

6 national rule overlays running on one shared engine

85 guardrails in force, 48 of them blocking rather than advisory

0.00% guardrail violation rate across the audited corpus

100% of decisions carry a named legal instrument

0 dispositions produced by a language model

Audited on 643 records in wave one of an eight wave, 5,304 record build.

WHO THIS IS FOR

Ministers, undersecretaries and authority leadership deciding whether automated disposition belongs in national food infrastructure, and on what terms.

The obvious answer is the wrong one

A single regional model trained on all six regimes will produce the average of six legal systems. The average is nobody's law. It would be wrong in five countries and right in one by accident, and no country could adopt it without ceding regulatory authorship to a training set.

What this paper proposes instead

Separate the two things that were wrongly fused. **Physics is shared. Law is not.** One engine holds the parts that are the same everywhere: what a temperature reading means, when a chain is broken, what counts as evidence. Six national overlays hold the parts that are not: who has jurisdiction, what must be notified, what may be moved, what may be said about a penalty. The overlay always wins.

Nothing about a country's law has to change for it to run. Each state keeps its own instruments, its own thresholds and its own escalation path. What the six share is the machinery, the audit format, and the standard of proof.

THE PROBLEM

Six rulebooks, one supply chain

A container does not stop being the same container when it crosses a border. The rules governing it stop being the same rules.

A load of chilled seafood leaves a supplier in South Asia. It clears at Jebel Ali, moves overland to Dammam, and part of it continues to Doha. Inside seventy-two hours it has passed under three separate competent authorities. Each of them has a different answer to the same question, and the differences are not cosmetic.

Kuwait	Moving a seized consignment without prior permission carries KWD 2,000 to 5,000 under MR 6/2023 Article 64(2). The instinctive operational response, move it somewhere colder, is the one that is penalised.
Saudi Arabia	No imported consignment releases before SFDA approval through FASEH, however clean the temperature record looks. Halal certification counts only from a body the SFDA Halal Center recognises.
Oman	Seafood falls to the Fish Quality Control Centre under Ministerial Decree 12/2009, not general food control. One excursion on an export load can reach the country's EU and US market access arrangements, not just the cargo.
United Arab Emirates	The governing local authority changes during the journey. Free zones are expressly inside scope under Article 19, which is where most cross-border assumptions break.
Qatar	Jurisdiction is shared between agencies, so notification requires a routing decision before any action is taken, not after.
Bahrain	The rejection procedure is the most prescriptive in the region, specified down to the branch of process, the request letters and the disposal site.

These are not differences of tone or emphasis. They are differences in what is legal. A system that smooths them into one regional answer is not harmonisation. It is six compliance failures wearing one interface.

What the gap costs before anyone argues about law

12%
of all food produced worldwide is lost because refrigeration was inadequate, not because the food was unsafe to begin with.

20 to 25%
is the loss range during transport specifically, driven by temperature swings and delays rather than spoilage at source.

Up to 85%
of Gulf food is imported, so almost every one of those losses lands inside a national border rather than at a farm.

ARCHITECTURE

The engine, end to end

Six layers between an artifact arriving from the field and a consignment being released. Every layer can stop the process. None of them can be skipped.

01 FIELD INTAKE
Logger CSV · Chat message · QC form OCR · Voice note

Whatever the port actually produced, not a clean form.

02 TELEMETRY QUALITY GATE
Is there enough evidence here to decide anything?

Runs before any judgement. Always.

03 RULE ENGINE · deterministic core
Temperature bands · excursion minutes · shelf life

Plain code. No model. Same answer every time.

04 GUARDRAIL OVERLAY · 85 rules
25 shared GCC + 10 per state · 48 blocking

Base loads first. The national overlay wins.

05 VOTING GATE
Five independent votes · disagreement escalates

Only for what a threshold cannot settle.

06 PROVENANCE ENVELOPE + DECISION TRACE
17 sealed fields · citation · vote record

Written before anything is released.

ACCEPT | HOLD FOR QA | REJECT | INSUFFICIENT DATA

Shared everywhere

Layers one to three. Temperature physics, evidence quality and the disposition logic itself. These do not vary by country because the science does not.

National, always

Layer four. Each state's overlay loads on top of the shared base and overrides it wherever the two differ. A country changes its own file and nothing else in the system moves.

Proof, not opinion

Layers five and six. What the system cannot settle by rule goes to a panel, and everything it decides is sealed with the conditions under which it decided.

HOW A DECISION IS MADE

From sensor to signature

The same six steps run on every consignment, in the same order, whether the artifact is a clean sensor export or a voice note from a driver.

1	2	3	4	5	6
ARRIVE	**QUALIFY**	**DECIDE**	**LOCALISE**	**VOTE**	**SEAL**
Field artifact is taken in as is	Truncation, sentinels, device ID collisions	Rule engine assigns one of four outcomes	National overlay applied, overlay wins	Panel of five on what rules cannot settle	Envelope and trace written, then release

1 Arrive
The system takes the artifact in the form the field actually produced it. A logger export, a chat message, a photographed quality form, a recorded note. Requiring a clean input is how most deployments quietly fail.

2 Qualify
Before anything is judged, the record is tested for whether it can support a judgement at all. Truncated logs, error sentinels sitting in a temperature column, two consignments sharing one device identifier.

3 Decide
The rule engine assigns one of four outcomes. It is ordinary code and it reaches the same outcome every time it sees the same evidence.

4 Localise
The national overlay is applied. Where the shared base and the country rule disagree, the country rule governs, without exception.

5 Vote
Anything a threshold cannot settle is put to five independent assessments. Low agreement escalates to a stricter review rather than being averaged away.

6 Seal
The envelope and the trace are written before release, not after. A decision that cannot be sealed is not a decision the system will issue.

The same six steps on one real record

What arrived

A driver's message and a logger export for a seafood load bound for Oman. The message says the readings were mostly fine.

What the system found

The word mostly is doing a lot of work. The export ends mid-line and covers nine hours of a chain the narrative says ran seven days.

What it issued

Hold for quality review, with the coverage gap named, the Omani instrument cited, and five assessments agreeing.

No step in that sequence required anyone to read a regulation at three in the morning.

COMPONENT ONE

Why a machine decides, and not a model

The most consequential decision in this architecture is a restriction: **no disposition in this system is ever produced by a language model.**

The rule engine is ordinary software. It reads its temperature bands, its allowed excursion minutes and its shelf-life limits from the guardrail pack, which in turn compiles them from GSO 150-1 and GSO 150-2 and from the six national law profiles. It returns one of four answers. It returns the same answer every time it sees the same evidence, and any engineer or auditor can re-run it line by line and get the same result.

This matters more to a regulator than to an engineer. An inspector can be asked to justify a decision. A rule engine can be handed over in full. A language model can do neither, because it cannot promise it will answer the same way next quarter, and an authority cannot enforce against something that will not hold still.

THE ONLY FOUR OUTCOMES

ACCEPT
Evidence supports release under the governing regime.

HOLD FOR QA
Something is unresolved. A qualified human decides.

REJECT
The chain is broken beyond the applicable limit.

INSUFFICIENT DATA
The record cannot support any conclusion.

AND ONE MORE

Abstain is always available to the system and is never counted against it. A machine that is allowed to say it does not know is worth more than one that is rewarded for always answering.

One rule shows the discipline

There is an instruction the industry knows well: release the warm load anyway, because the delivery window matters. The system keeps that instruction in its vocabulary for one reason only, so that it can recognise it and refuse it. The engine never issues it. Every path that once led there now leads to hold for quality review instead.

Commercial pressure does not convert an excursion into a release. That sentence is not a policy statement in a manual. It is enforced in code, and the audit confirms it held on every record.

What an auditor can do here that they cannot do today

Reproduce it

Run the engine on the same record and get the identical outcome.

Diff it

Compare this quarter's rules against last quarter's and see exactly what changed.

Challenge it

Point at the specific threshold they dispute rather than at the system as a whole.

COMPONENT TWO

The national overlay

Eighty-five rules across seven files. Twenty-five are shared across the Gulf. Ten belong to each state. Forty-eight of them stop a decision rather than annotate it.

Load order is the entire design. The shared base loads first. The country overlay loads second. **Where they disagree, the country wins.** A state that amends its regulation edits one file and the system changes behaviour in that state alone.

UAE	Saudi Arabia	Qatar	Kuwait	Oman	Bahrain
10 rules	10 rules	10 rules	10 rules	10 rules	10 rules

SHARED GCC BASE — 25 rules · loaded first · overridden by any national rule above

Where the rules came from

Not from a workshop. From ten records that the pipeline had already thrown away.

All ten were synthetic and every one had failed an earlier check. A system that quietly issued a disposition on them would have been wrong ten times out of ten, and it would have been wrong before reading a single temperature. That finding shaped the pack more than the physics did, and it is why the first question the system asks is never about the cargo. It is about whether the record deserves to be believed.

WHAT THE RULE STOPS	WHY IT WAS WRITTEN
Deciding on unvalidated data	Provenance is checked before evidence. This rule cannot be overridden by an instruction written inside the artifact.
Reading a cut-off log as final	Every logger export in the failure sample ended mid-line. The last complete row was not the last reading.
Averaging in an error code	A sensor fault value of minus 99.9 sitting in a temperature column makes a warm chain look cold.
Trusting the file name	Metadata, filenames and scenario labels are not evidence. Two records leaked their own answer through a header.
Quoting a fine that is not sourced	Where a national penalty could not be verified, the rule forbids stating any figure and requires escalation.

That last row is the one to remember. A plausible fabricated fine is more dangerous than an admitted gap, because it will be acted on.

COMPONENT THREE

No single opinion releases a shipment

Some questions cannot be reduced to a threshold. For those, the system does not ask once. It asks five times, independently, and counts.

Whether a field narrative is genuine or invented. Whether an abstention was the right call or an evasion. Whether a claim in the text is supported by anything in the record. These are judgement calls, and a judgement call from a single source is an opinion, not a finding.

When agreement falls below the floor, the case is not averaged out. It escalates to a stricter review. Every vote and every stated reason is written down, so an auditor opening the file a year later can see exactly where the panel split and why.

GATE A	Before the data is used
Schema validity	at or above 99.5%
Round trip recovery	at or above 95%
Near duplicate rate	at or below 3%
Label leakage probe	at or below 70%
Guardrail violations	at or below 1%

PASS →

GATE B	After the model is trained
Malformed output	at or below 0.5%
Invented fields	at or below 1%
Abstention precision	at or above 85%
Cross jurisdiction gap	at or below 5 points
Adversarial gap	at or below 10 points

ANY CHECK FAILS · THE WAVE HALTS · NOTHING SHIPS ON A PARTIAL PASS

Two gates, in sequence

Gate A runs on the evidence before it is used for anything. Gate B runs on the system's behaviour after it has been tuned. Either one can stop the process on its own, and a partial pass is treated as a failure.

The check that carries the promise

Cross jurisdiction gap. Accuracy between any two of the six states may not differ by more than five points. A system that performs well in Dubai and poorly in Muscat fails and does not ship. This is where harmonisation stops being a claim and becomes a number someone can audit.

What happens when the panel disagrees

1. Disagreement is recorded, not resolved by majority.
2. The case is re-put under a stricter standard, with the split visible.
3. If it still will not settle, it leaves the machine and goes to a person.
4. Every vote and stated reason is retained, whichever way it ended.

A disagreement that is averaged away is a disagreement that will resurface in a dispute, months later, with nobody able to explain it.

COMPONENT FOUR

Every decision arrives sealed

An envelope travels with each record. It captures not only what was decided, but the exact conditions under which it was decided.

IDENTITY	EVIDENCE	MACHINERY	REVIEW
`state_id`	`artifact_type`	`rule_engine_sha`	`screener_verdict`
`wave`	`language`	`prompt_template_hash`	`round_trip_ok`
`cell`	`generator_model`	`rng_seed`	`human_reviewed`
`jurisdiction`	`renderer_model`	`is_adversarial`	`parent_request_id`

Most audit trails record the outcome. This one records the machine that produced it: which version of the rules was in force, which template was used, which random seed was drawn, whether a human ever looked at it, and whether the record survived its own consistency check.

Why a minister should care about seventeen fields

Because it converts a technical detail into a recall boundary.

Suppose a temperature threshold is revised next year, or a rule is found to have been applied too loosely. The question an authority has to answer within hours is: which consignments were released under the old rule, and where are they now.

With this envelope, that is a single query. Without it, the honest answer in most operations today is that nobody can draw the line, so the recall is either far wider than it needs to be or far narrower than it should be. Both outcomes are expensive. Only one of them is dangerous.

WHAT THE SEAL ANSWERS

1. Which version of the rules decided this, exactly
2. Whether a human ever reviewed it
3. Whether the record passed its own consistency check
4. Which jurisdiction's overlay was in force

The envelope is written before release. A decision that cannot be sealed is not issued.

A worked case: the threshold moves

Monday

A national standard revises the allowed excursion window for chilled seafood downward. The change takes effect immediately.

Within the hour

Every consignment released under the previous window is listed by rule version, with its port, date and current holder.

Same day

The authority issues a notice covering exactly that set. Not the whole month's imports, and not a guess.

COMPONENT FIVE

What the regulator actually sees

Everything described so far exists to make one screen honest. The trace is the product. The rest is machinery.

DECISION TRACE	ASSET-771904 / wave 01 / OM
CONSIGNMENT	Chilled finfish and seafood · export consignment
COMPETENT AUTHORITY	Fish Quality Control Centre, Sultanate of Oman
DISPOSITION	**HOLD FOR QA** not releasable without authority contact
EVIDENCE READ	`36 readings, 0.80 to 3.03 C, window 9h 00m`
FLAG RAISED	Log covers 4.5% of the stated 7 day chain
RULE FIRED	Coverage insufficient for the claimed period · BLOCKING
LEGAL BASIS	Ministerial Decree 12/2009 · verify at source
PANEL	5 votes cast · 5 in agreement · no escalation
SEALED UNDER	`rules 1.1.0 · engine 338973fdc339 · seed 165423786309575`

Every line above is produced by the system, not written by an operator.

Read it as a document, not a dashboard

There is no confidence percentage on this screen and no coloured gauge, because neither survives a dispute. What survives is the reading that was taken, the gap that was found, the rule that fired, and the instrument that rule sits under.

An importer who disagrees is not arguing with a machine. They are pointing at a specific line and naming what they think is wrong with it. That is a far better argument to have.

What the trace deliberately will not do

It does not state a penalty amount unless that amount was verified at source. It does not interpret the instrument it cites. It names it, so a lawyer in the competent authority can go and read it.

The system is built to be checkable, not to be believed.

The test to apply to any system that claims to do this

Ask the vendor to show you a single decision, in full, including the one it got wrong. If the answer is a confidence score, a heat map or a dashboard, the system cannot be audited. If it looks like the panel above, it can be argued with, and that is the only property a regulator needs.

EVIDENCE

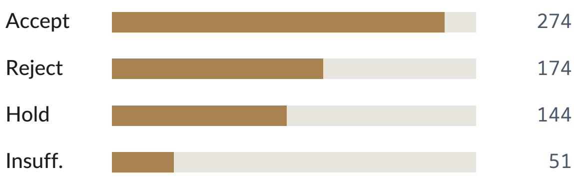
What the reference build produced

One wave of a planned eight. Six hundred and forty three records, audited rule by rule, with the results published rather than summarised.

0.00% guardrail violation rate	**100%** schema validity	**100%** round trip recovery	**0.991** mean round trip confidence	**0** autonomous warm releases

RECORDS AUDITED BY JURISDICTION

OUTCOMES ISSUED



RECORDS BY SOURCE TYPE

What these numbers show

Coverage is even. No jurisdiction is over-represented and no single outcome dominates. That balance is what makes the cross jurisdiction check on the previous page meaningful rather than decorative.

The zero that matters most is the last one. Not a single record was released warm under commercial pressure, because the path does not exist.

What they do not show

This is a reference build on simulated physics, not a live deployment. The thresholds are working defaults and still need sign-off from domain experts against each country's technical regulations. Several penalty figures remain unverified and are deliberately left unstated.

What is proven here is the architecture, not the clinical thresholds. Anyone presenting it as more than that should be asked harder questions.

What has to happen before this touches a real consignment

Expert sign-off

Each country's technical regulator confirms the temperature bands and excursion limits used for its own overlay.

Source verification

The penalty figures currently marked unstated are either confirmed at source or stay unstated permanently.

Live comparison

The engine runs beside human inspection long enough to show where the two diverge.

CONCLUSION

What this changes for national food infrastructure

1

From capacity to coverage

Today the number of consignments that get a proper regulatory judgement is limited by the number of qualified people available to make one. Under this design the judgement is made at the point of arrival, in seconds, with the same reasoning applied to the first container of the day and the four hundredth. Human expertise moves to the cases that actually need it.

2

From six systems to one that stays six

The six states can share the engine, the guardrail format and the standard of proof without any of them adopting another's law. That is the only form of regional harmonisation that does not require a treaty, because nothing sovereign is being surrendered. Each country still writes its own file.

3

From opinion to evidence

Every disposition arrives with the instrument that authorised it and the reading that triggered it. A dispute becomes a comparison of documents rather than a contest of assertions, and it resolves in hours instead of weeks.

What it does not do

- It does not replace the competent authority. It routes to it faster and with a complete file.
- It does not give legal advice. It names the instrument so that someone qualified can verify it.
- It does not learn from its own decisions. Ground truth is code, never consensus.

THE PROPOSED NEXT STEP

A ninety day pilot

One product category, one port, running alongside existing inspection rather than in place of it. The engine issues a disposition. The inspector issues theirs. Nothing is released on the engine's word.

The cost of the engine being wrong during that window is zero. What it buys is a precise map of where national law and machine reasoning diverge.

SCOPE	one product category, one port
RISK TO CARGO	none, the engine does not release
WHAT YOU GET	a divergence map and an audit format

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Full guardrail pack, knowledge base and audit output available for technical review on request.